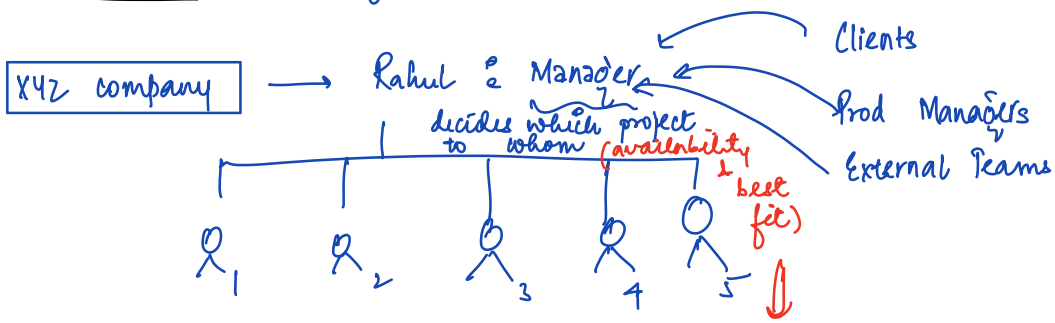
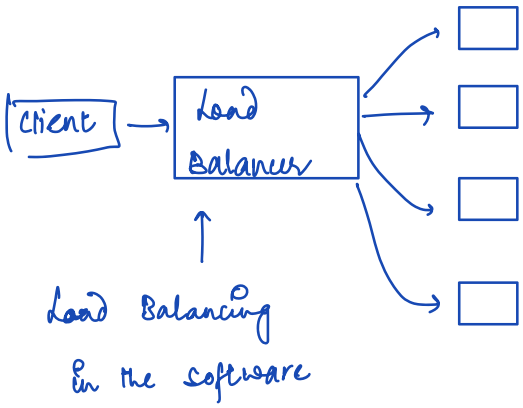
Load Balancers ~ Manager in an IT company



Everyone gets equal work – else some'll be overworked and others will be free

So, Rahul is doing load-balancing



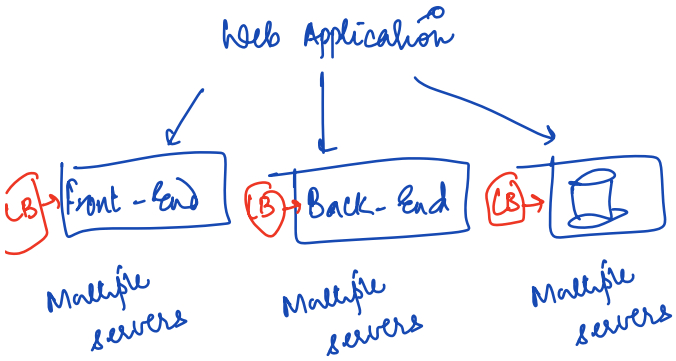
Things that Load Balancers do –

(Manager – everyone is equally worked
→ Leave/Unavailable who?
→ Manage work – so that its always on line

1. Each server is equally loaded
2. Health check of every server
3. High scalability / availability / High Throughput (Requests are sent across servers)

What all places where LBs are put?

Manager needed when there are many individual components (2Cs) to manage them



(If we horizontally scale, then we can put LB there)

When to use load balancers?

- Multiple app-instances / servers (else difficult for client to remember all those servers)

Eg if there is no Rahul, then external client has to talk to every other person.

When not to use?

- Monolithic system - everything running on single m/c - we don't need load balancer.

[Vertical Scaled]

Benefits (as we're adding extra layer, we should know it)

- Better customer experience
  (high availability)
  (achieve scalability)
  (Faster response time - server free)
- Minimise downtime.
- Filtering / Understanding request patterns
  ↳ Prediction of traffic bottlenecks.
- Single server not overloaded

↳ Manager ① external client / prod. manager as one point of contact - better customer experience.
② Product will be managed.
③ Understands customers requirements, filtering of certain requests.
④ Single employee not stressed.

Advantages of using load balancers. 1. Optimisation: Load Balancers help in resource utilisation and

lower response time

, thereby optimising the system in a high traffic environment. 2. Better User Experience: Load balancers help in reducing the latency and

increasing availability

, making the user's request go smoothly and error-free. 3.

Prevents Downtime

: Load Balancers maintain a record of servers that are non-operational and distribute the traffic accordingly, therefore ensuring security and preventing downtime, which also helps increase profits and productivity. 4. Flexibility: Load Balancers have the flexibility to re-route the traffic in case of any breakdown and work on server maintenance to ensure efficiency. This helps in avoiding traffic

bottlenecks

. 5. Scalability: When the traffic of a web application increases suddenly, load balancers can use physical or virtual servers to deliver the responses without any disruption. 6. Redundancy: Load Balancing also provides in-build redundancy by re-routing the traffic in case of any failure.

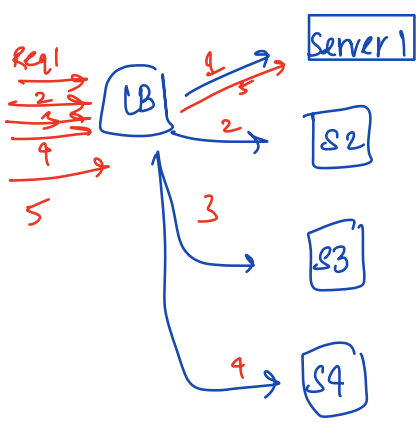
Load Balancing Algorithms

LB → Health check? if not → repair

→ Algorithm → to decide which server to use?

① Round Robin Method:

Load Balancer will choose each server one after the other.



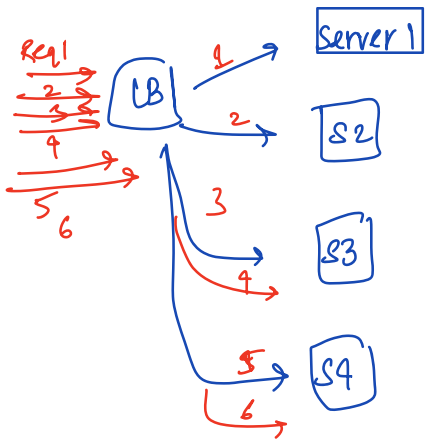
servers working one after other

→ Used when when servers are of equal capacity

Servers are allotted requests sequentially

② Weighted Round Robin

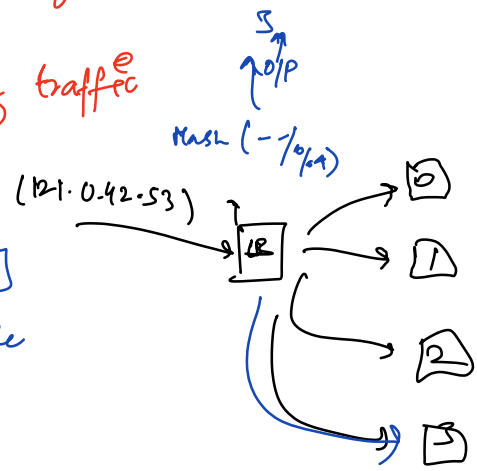
Used when servers are of diff^t capacity



③ IP Hash Algorithm → Hash function is applied on IP address of client

When Servers same capacity

used :- Random Unbiased distribution of traffic

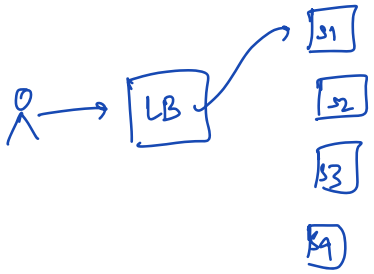


Hash (Request) % n] → [0, n-1]

Node

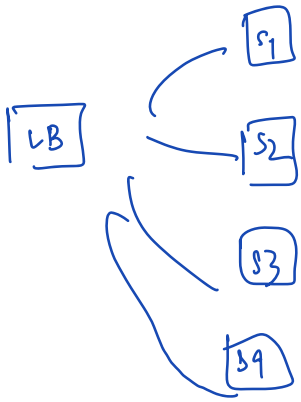
④ Least Connection Algorithm

Used when : We need to decide based on availability of open connections on a server.



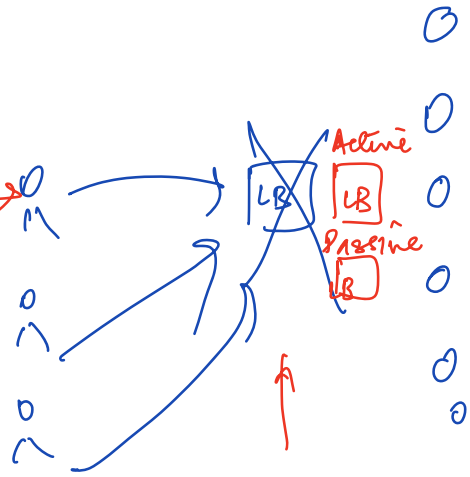
5) Least Response Time

When → Client is looking for least response time



Challenges

Can be single point of failure



Single pt of failure

[Anti-Pattern]

Bring passive redundancy here.